

Kevin González Ortega

Data Scientist | Psychometrics and Educational Measurement | Machine Learning and AI Evaluation
Bloomington, IN, USA | kevingpanama@gmail.com | linkedin.com/in/kevingpanama |
ORCID: 0000-0003-0437-5136

SUMMARY

Data scientist and Systems and Computing Engineer training in psychometrics and educational measurement as a Fulbright Scholar in the M.S.Ed. in Qualitative and Quantitative Research Methodology at Indiana University Bloomington. Ten years at Fundación Ayudinga, a nonprofit e-learning foundation, where I authored its first R&D&I plan, direct a portfolio of 10+ projects in machine learning, knowledge management and gamification for education, designed the data governance strategy, and grew the volunteer program from 6 to 42 members. Research assistant on a multi-institution text-mining project with Universidad Tecnológica de Panamá and McGill University's PRISM Consortium, with applied work in NLP, large language models and geospatial data; undergraduate thesis on knowledge graphs for personalized learning pathways, recognized as outstanding student research. Co-author of two peer-reviewed conference papers: an IEEE paper on data governance for nonprofit educational organizations and an APANAC paper on geospatial health and nutrition data. Taught seven undergraduate courses in mathematics, discrete structures, web programming and artificial intelligence for video games. I combine measurement, statistical modeling and machine learning to build and evaluate assessments and AI systems that are valid, reliable and fair.

RESEARCH INTERESTS

- Educational measurement and item response theory, including methods that estimate a student's proficiency with the fewest possible items.
- Validity and fairness of assessments and algorithmic systems; evaluation of AI and LLM-based tools used in education.
- Diagnosing learning gaps by combining item response theory with knowledge graphs; personalized learning pathways and learning analytics.
- Machine learning for education and the social sciences, including student motivation and gamification in virtual learning environments.
- Standardized assessment of student competencies to inform data- and evidence-based education policy.

EDUCATION

Indiana University Bloomington, Bloomington, IN, USA

Master of Science in Education (M.S.Ed.), Qualitative and Quantitative Research Methodology | Aug 2025 – May 2027 (expected)

- Fulbright Scholar. Emphasis in measurement, multivariate statistics and machine learning for the social sciences.
- Completed coursework: Intermediate Statistics Applied to Education; Multivariate Analysis in Educational Research; Strategies for Educational Inquiry; Qualitative Inquiry in Education; Computer Laboratory and Consultation for Educational Inquiry; Network Science; Education Policy and Reform.
- Current coursework (Fall 2026): Educational Assessment and Psychological Measurement; Machine Learning Methods for Education and Social Science; Applied Missing Data Analysis.
- Planned (Spring 2027): Longitudinal Data Analysis; Master's Thesis in Education.

University of North Texas, Intensive English Language Institute, Denton, TX, USA

Fulbright Pre-Academic English Program | Jan 2025 – Jul 2025

Universidad Tecnológica de Panamá, Panama City, Panama

Bachelor of Engineering, Systems and Computing Engineering | Mar 2016 – Dec 2023 (degree conferred Jan 2024)

- Thesis: Design and Implementation of Knowledge Graphs for Building Personalized Learning Pathways for Fundación Ayudinga (2023); recognized as outstanding student research.
- Relevant coursework: Calculus I–III; Ordinary Differential Equations; Advanced Engineering Mathematics; Numerical Methods for Engineering; Statistics; Discrete Structures; Artificial Intelligence; Applied AI Tools; Knowledge-Based Systems; Databases I–II; Modeling and Simulation; Research Methodology in Engineering.
- Organizer, Symposium on Emerging and Disruptive ICT for Social Development (STEEDS).

FELLOWSHIPS AND HONORS

Fulbright Faculty Development Program Scholar, U.S. Department of State | Jul 2025

Awarded to pursue graduate study in research methodology at Indiana University Bloomington, preceded by an intensive English program at the University of North Texas.

Outstanding Student Recognition for Research, Universidad Tecnológica de Panamá | Dec 2021

Recognized for undergraduate research on knowledge graphs for generating personalized learning pathways.

Fellow, Latin American Citizen Action Laboratory 2.0, Jóvenes Unidos por la Educación and Panama Canal Authority | 2020

Fellow, International Citizen Advocacy Laboratory, Educación 2020 and Jóvenes Unidos por la Educación | 2018
Chile–Panama cooperation initiative on education policy. Project recognized among the top three; awarded a public-policy internship in Chile.

RESEARCH EXPERIENCE

Universidad Tecnológica de Panamá, Faculty of Computer Systems Engineering (FISC), Panama City, Panama
Research Assistant, Text Mining on Reported Nutrition Interventions in Panama | Oct 2022 – Jun 2023

- Supported research tasks in a text-mining project led by Principal Investigator Javier Sánchez-Galán, PhD, conducted by UTP-FISC with the PRISM Consortium (McGill University), Panama's National Research System (SNI, SENACYT) and CEMCIT-AIP.
- Co-authored with the PI a 2023 APANAC conference paper (presented as a poster) on a geospatial information system for food-security, nutrition-intervention and human-health data in Panama (see Publications).

Natural Language Processing and LLM Project, National Research System (SNI), Panama | Oct 2022 – Jul 2023

- Built an ETL pipeline to generate a text corpus, then cleaned the corpus and measured its quality.
- Implemented text-embedding algorithms and visualized the embeddings with dimensionality-reduction techniques.
- Developed a chatbot-style system powered by large language models (LLMs).

Geospatial Data Visualization, CEMCIT-AIP, Panama | Oct 2022 – Dec 2022

- Developed semi-automated extraction of tabular data and georeferenced health and nutrition indicators.
- Built a dashboard for consulting and comparing the indicators.

Undergraduate Thesis: Knowledge Graphs for Personalized Learning Pathways, Universidad Tecnológica de Panamá | 2023

- Designed and implemented knowledge graphs to represent knowledge and build personalized learning pathways for Fundación Ayudinga.
- Combined AI algorithms with knowledge-graph representations to reduce learning deficiencies, a central post-pandemic problem, with an approach designed to support diagnosis of deficiencies in any field of study.
- Earned an outstanding-student recognition from the university for this research.

PROFESSIONAL EXPERIENCE

Fundación Ayudinga, Panama

Director of Technology and Educational Research | Jun 2019 – Present

- Authored the foundation's first R&D&I plan and direct its portfolio of 10+ research projects in machine learning, knowledge management and gamification applied to education.
- Lead research on factors that influence learning, analyzing experimental data to build and evaluate artificial intelligence models. Research lines include student motivation in virtual learning environments, personalized teaching and learning, gamification as a motivational strategy, and LLMs and knowledge graphs for knowledge representation in educational organizations.
- Designed the department's organizational structure, roles, teams and processes and reviewed its training programs, growing the volunteer program from 6 to 42 members.
- Designed and implemented the data governance strategy (2021–2023): roles, processes and policies for documentation, privacy, metadata, storage, data quality and business intelligence; a data architecture with high-availability and fault-tolerance mechanisms; and data-literacy training for volunteers. Co-authored an IEEE CONCAPAN 2025 paper on data governance for nonprofit educational organizations (see Publications).
- Led the design and implementation of the development project for the foundation's e-learning platform; designed

and maintain its cloud infrastructure, cloud services and web servers, and lead systems analysis and design for web and desktop applications.

- Contributed to the foundation's COVID-19 response, which began streaming and broadcasting mathematics and science classes on open television to Panama and the region the day after schools closed.

Director of Finance and Treasurer of the Board | Feb 2020 – Present

- Built the foundation's finance and administrative function from scratch, consolidating its accounting information.
- Automated the collection of financial information for decision-making; prepare and manage the annual budget.

Backend Lead | Aug 2017 – Jun 2019

- Designed the microservices architecture of the e-learning platform, improving scalability and development flexibility.
- Led the backend development team using agile methods and ensured integration across microservices.

Junior Software Developer | Feb 2016 – Jul 2017

- Designed, developed and launched a pendulum simulator in video-game format (Unity 3D) that lets learners explore the physics of the pendulum interactively.
- Developed features for the foundation's e-learning platform.

TEACHING EXPERIENCE

Universidad Interamericana de Puerto Rico, Panama Campus, City of Knowledge, Panama City, Panama

Adjunct Professor (Part-time) | Aug 2023 – Dec 2024

- Taught seven undergraduate courses (nine sections) in mathematics and computer science:
 - GEMA 1200 Fundamentals of Algebra (Spring 2024, Fall 2024)
 - MATH 1500 Precalculus (Fall 2023, Fall 2024)
 - MATH 2251 Calculus I (Spring 2024)
 - COMP 2501 Discrete Computer Structures I (Fall 2023)
 - COMP 2502 Discrete Computer Structures II (Spring 2024)
 - COMP 3015 Web Programming with Databases (Fall 2024)
 - GAME 3400 Artificial Intelligence for Video Games (Fall 2024)
- Developed course materials and delivered lectures, practice sessions and tutorials.
- Assessed student progress through exams, assignments and projects, with individualized feedback and academic advising.

Tutorez, Panama City, Panama

Tutor, Knowledge Heroes (Héroes del Conocimiento) Free Tutoring Program | Apr 2023 – Jul 2023

- Tutored public-school students from across Panama in mathematics, physics and chemistry through daily individual sessions for three consecutive months; designed lessons and interactive labs and provided personalized follow-up for each student.
- Previously delivered 71 hours of private tutoring to Tutorez clients (collaboration since Aug 2018).

UTP-FISC Tech Academy, Universidad Tecnológica de Panamá, Panama City, Panama

Volunteer Instructor | Feb 2018 – Mar 2020

- Designed, planned and taught Python for Programmers (2020), Introduction to Python (2019) and NoSQL Databases (2018).

RESEARCH AND PROJECTS

Impact Data Analysis, Safe Success Education Project, Blue Green Foundation | Apr 2019 – Jun 2019

- Compiled and integrated impact-indicator data and conducted a descriptive analysis of five years of program results; prepared the final report and presented the findings.

Gamified, Adaptive Learning Tool for Fractions, Fundación Ayudinga

Research Advisor

- Co-advised, with Johel Batista, a student team that designed a fractions-learning tool combining gamification, cognitive training, spaced repetition and adaptive learning, informed by working memory and math anxiety; presented at the Jornada de Iniciación Científica (JIC), Universidad Tecnológica de Panamá.

Cambio en 10, International Citizen Advocacy Laboratory | Mar 2018 – Aug 2018

- School-dropout prevention project built on life-skills development, with emotional intelligence as a cross-cutting element of comprehensive education.

PUBLICATIONS

Peer-Reviewed Conference Papers

Verbel, G., **Gonzalez, K.**, Muñoz, A. C., Lopez, V., & Sanchez-Galan, J. E. (2025). Democratizing data analytics in education: A governance strategy for non-profit educational organizations. In *2025 IEEE 43rd Central America and Panama Convention (CONCAPAN XLIII)*. IEEE. <https://doi.org/10.1109/CONCAPAN66820.2025.11512483> [PDF]

González Ortega, K., Aguilar, E., Aizprúa, A. G., Cedeño, E., & Sánchez-Galán, J. (2023). Estructuración de un sistema de información geoespacial para el análisis de datos de seguridad alimentaria, intervenciones nutricionales y de salud humana en Panamá [Structuring a geospatial information system for the analysis of food security, nutrition intervention and human health data in Panama]. In *XIX Congreso Nacional de Ciencia y Tecnología (APANAC 2023)* (pp. 356–362). APANAC. Presented as a poster. <https://doi.org/10.33412/apanac.2023.3959> [PDF]

Undergraduate Thesis

González Ortega, K. (2023). *Diseño e implementación de grafos de conocimientos para la construcción de rutas de aprendizajes personalizadas para la Fundación Ayudinga* [Design and implementation of knowledge graphs for building personalized learning pathways for Fundación Ayudinga] [Undergraduate thesis]. Universidad Tecnológica de Panamá.

LEADERSHIP AND SERVICE

Jóvenes Unidos por la Educación, Panama

Selection and Recruitment Coordinator | Nov 2018 – Aug 2020

- Built the applicant selection and recruitment system for the Latin American Citizen Action Laboratory and its interactive virtual platform.

Operations Team Member | Nov 2016 – Nov 2018

- Co-organized #HablemosDeEducación19 (Aug 2018 – May 2019), a citizen advocacy campaign that drafted 19 education policy proposals for Panama's 2019 general elections, presented and discussed them with the presidential candidates, and set out to measure their execution by the next government.
- Co-organized the first and second editions of "The Role of Youth in the Future of the Panama Canal," held with the Panama Canal Authority, in which 400+ young people presented their views on the Canal's challenges and future.

Laboratorio Latinoamericano de Acción Ciudadana, Panama

Mentor, AC7IVATE! | Jun 2019 – Dec 2021

- Mentored a socio-emotional learning program for students aged 11–12 built around six challenges: self-esteem, recognizing and accepting emotions, assertive communication, teamwork, conflict resolution and critical thinking.

CERTIFICATIONS

- International Diploma in Data Protection Regulation in Panama, Universidad de Panamá, Apr 2025 (Credential ID VIEX-62932-D)
- Machine Learning with Big Data (2020); Big Data Integration and Processing (Nov 2019, Credential ID G75XSE3CL7A2); Big Data Modeling and Management Systems (2019); Introduction to Big Data (2019), University of California San Diego (Coursera)
- The Data Scientist's Toolbox (2020); R Programming (2020), Johns Hopkins University (Coursera)
- Object-Oriented Data Structures in C++ (Coursera), 2020

SKILLS

Programming and Data: Python, R, SQL, NoSQL, JavaScript, Java, C++, C#, MATLAB, .NET, HTML, Unity 3D, Dataiku DSS, Microsoft Excel

Statistics and Measurement (graduate coursework): Multivariate Analysis, Applied Intermediate Statistics, Educational Assessment and Psychological Measurement (in progress), Applied Missing Data Analysis (in progress), Network Science, Quantitative and Qualitative Research Design, Descriptive Impact Analysis

Machine Learning and AI: Machine Learning, Natural Language Processing (NLP), Text Embeddings, Dimensionality

Reduction, Large Language Models (LLM), Chatbots, Knowledge Graphs, Data Mining

Data Engineering and Management: ETL Pipelines, Data Governance, Data Quality, Metadata Management, Data Privacy Regulation, Business Intelligence and Dashboards, Big Data, Cloud Computing, Microservices, CI/CD, Geographic Information Systems (GIS)

Education and Management: University Teaching, Tutoring, Educational Technology, Learning Analytics, Agile Project Management, Budgeting

PROFESSIONAL MEMBERSHIPS

American Educational Research Association (AERA), Member | National Council on Measurement in Education (NCME), Member | IEEE, Member (Oct 2025 – Present) | Association for Computing Machinery (ACM), Member (Mar 2019 – Present)

LANGUAGES

Spanish (native) | English (professional; graduate study in English at Indiana University Bloomington)